

Laptop → Wifi Router — ISP — R1 — R2 — R3 — B

OSI (Open System Interconnection)

- 7) Application layer
- 6) Presentation layer
- 5) Session
- 4) Transport
- 3) Network
- 2) Data link
- 1) Physical

TCP/IP Mode

- | | | | |
|----|-------------|---|-----------------|
| 5) | Application | — | HTTP, DNS |
| 4) | Transport | — | TCP, UDP |
| 3) | Network | — | IP |
| 2) | Data link | — | Ethernet, Wi-Fi |
| 1) | Physical | — | |

Bit

→ Single binary digit ⇒ 0/1

Byte $\Rightarrow$ 8 bits

1000 bits $\Rightarrow$ 1 Kilobit (Kb)

1000 Bytes $\Rightarrow$ 1 KiloByte (KB)

1000 Kb $\Rightarrow$ 1 Megabit

1000 KB $\Rightarrow$ 1 MegaByte

100 Mbps $\Rightarrow \frac{100}{8} = 12.5 \text{ MB/s}$

Q How are bits transmitted.

01001000

App Layer



Bytes



Bits



Encoded into physical signals



Transmission medium

Transmission Rate

$$R = 100 \text{ Mbps}$$

$$d_{\text{trans}} = \frac{L}{R} \rightarrow \begin{array}{l} \text{length of packet} \\ \text{Transmission Rate} \end{array}$$

* Network operates on groups of bits

Bits



Frame



Contains IP packet



contains Transport data



contains Application Data

Application Layer

GET /index.html HTTP/1.1

Application Data



Transport layer

[TCP Header | Data] $\Rightarrow$ TCP Segment



Network layer

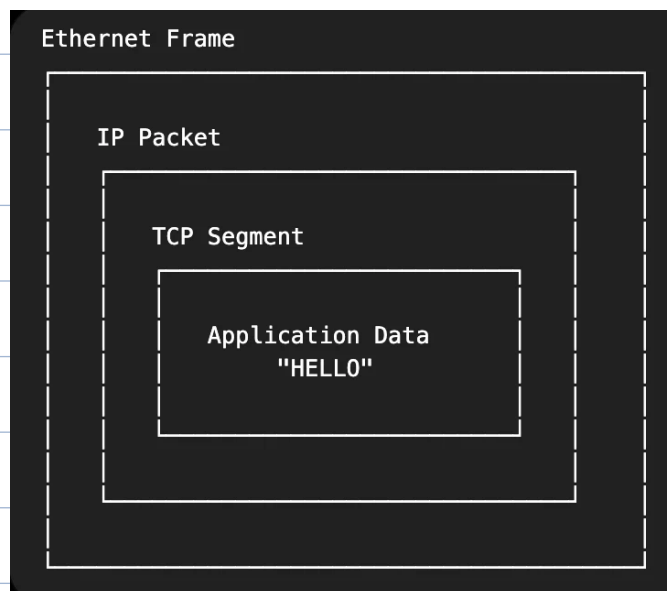
[IP Header | Payload] $\Rightarrow$ IP Packet



Data link layer

[Ethernet Header | IP Packet | Ethernet Trailer]

Ethernet frame



Payload.

Protocol Data Unit (PDU)

Layer

PDU

Application

Data / Message

Transport

Segment (TCP) /
Datagram (UDP)

Network

Packet / Datagrams

Data Link

Frame

Physical

Bits

Sender

Receiver

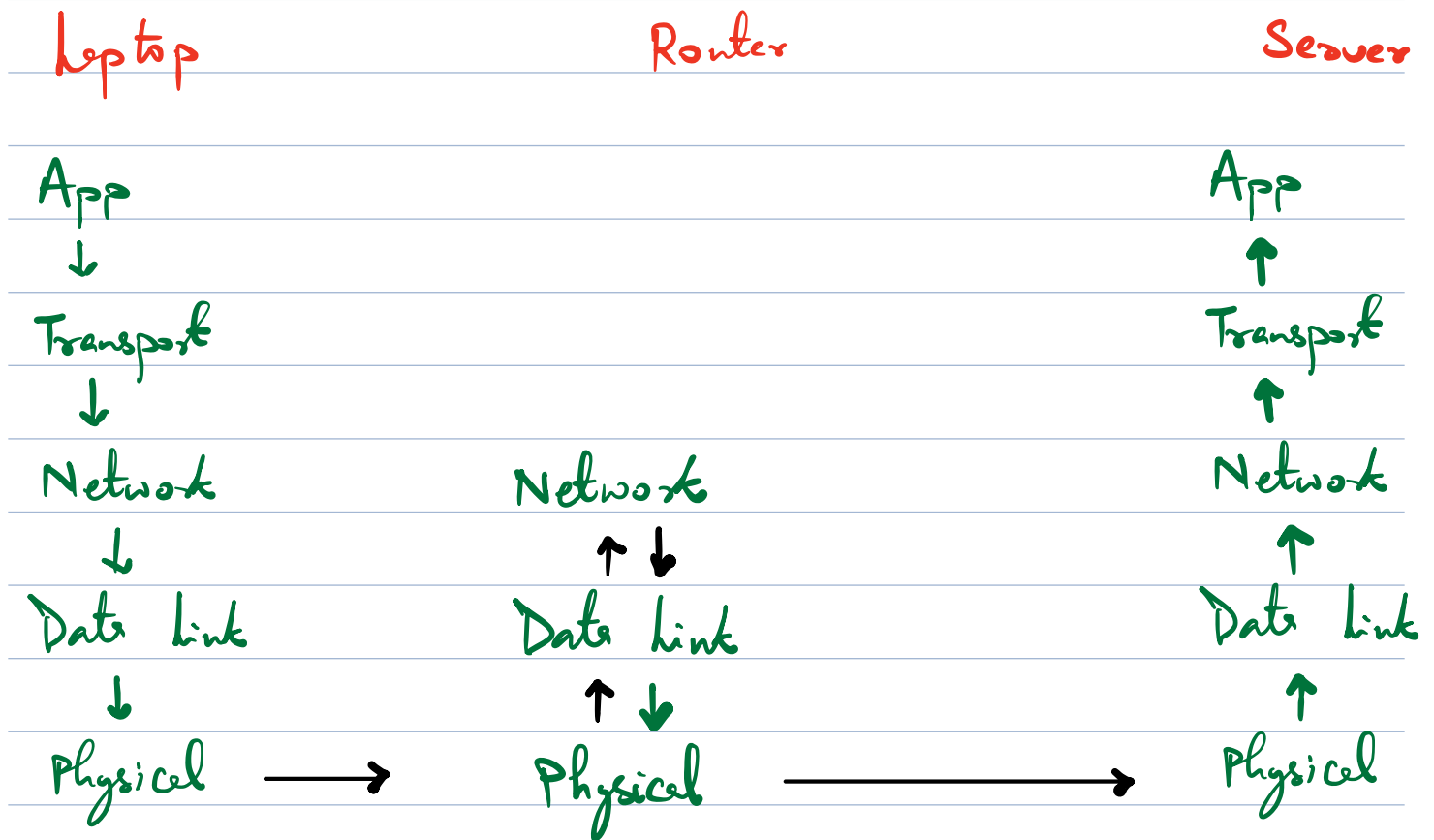
Application <-----> Application

Transport <-----> Transport

Network <-----> Network

Data Link <-----> Data Link

What Happens at a Router



Application layer

Application Data

GET /index.html



TCP Header

Source Port	Dest. Port	Seq / ACK	Data
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IP Header

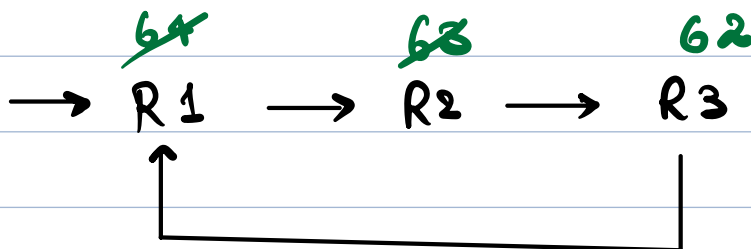
Source IP	Dest IP	Protocol	TCP Segment
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Upper Layer Protocol

[TCP
 [UDP
 [ICMP

TTL ⇒ Time to live



Ethernet Header

Destination MAC	Source MAC	EtherType	IP Packet
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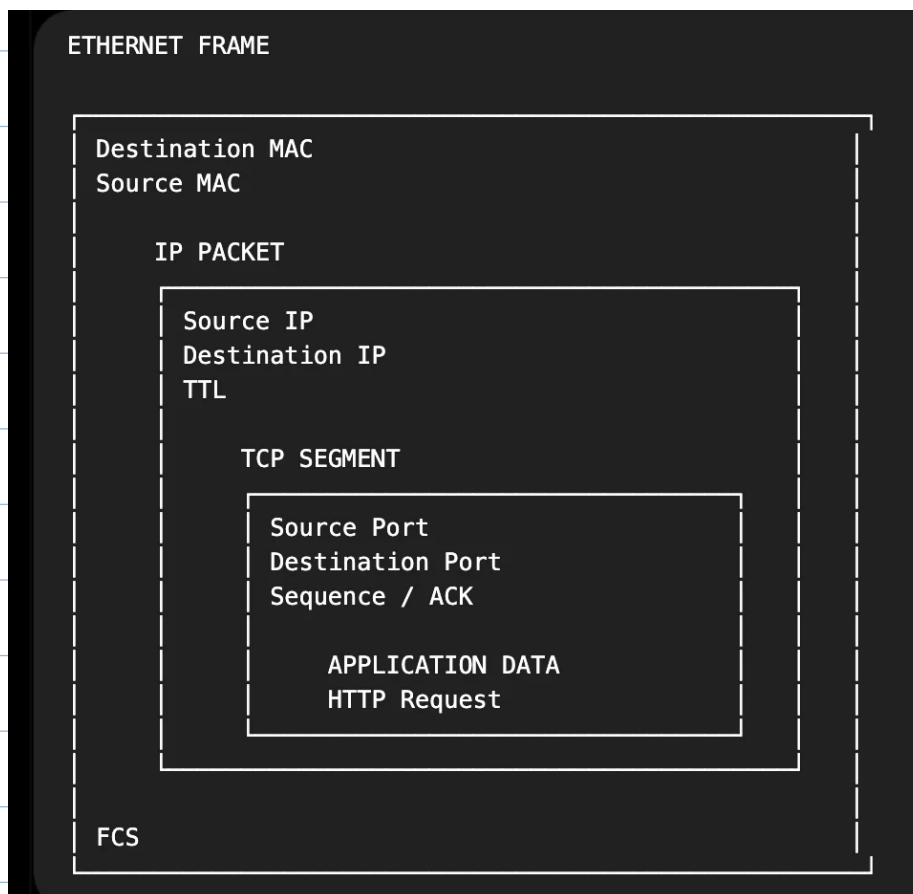


- IPv4
- IPv6
- ARP

Ethernet Trailer

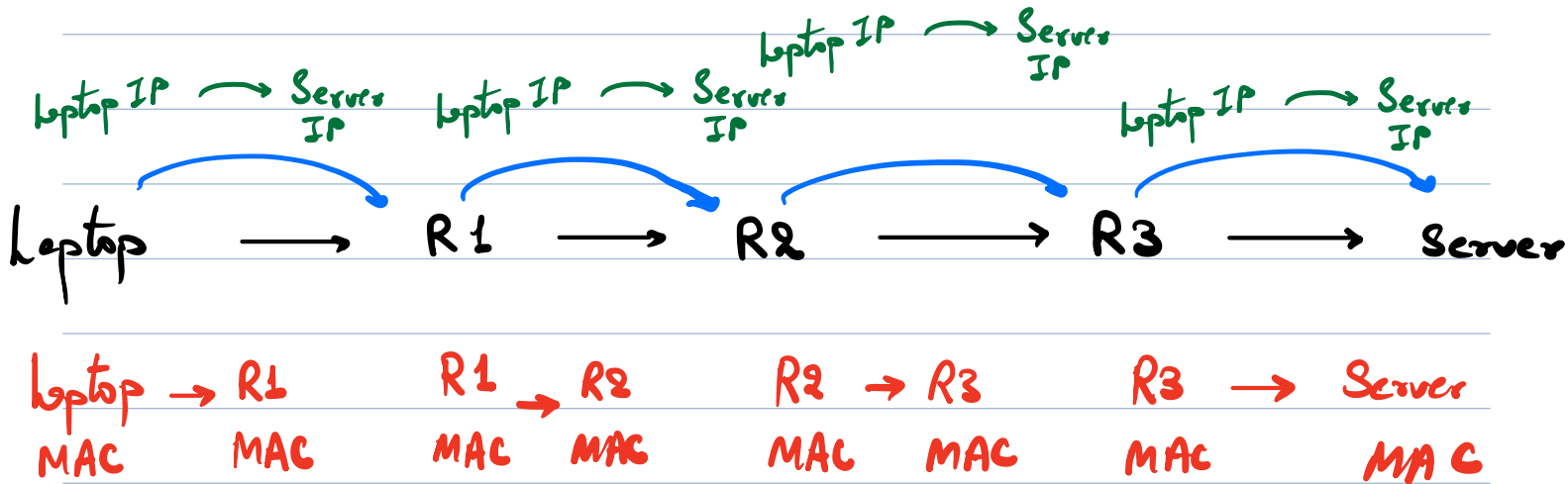
FCS → Frame Check Sequence

EtherNet Header	Payload	FCS
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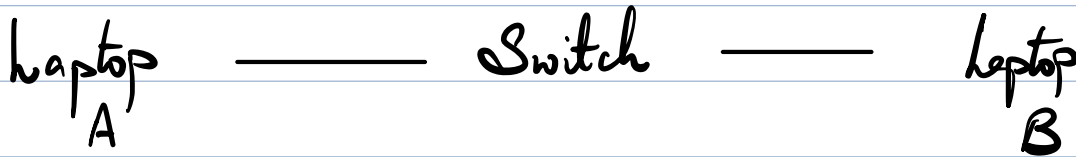


Types of Addressing

MAC	—	Link / Local delivery.
IP	—	Host / network-level addressing
Port	—	Process / Application end point.



Local Network



Ethernet

family of technologies widely used for wired local Area Network.

- Frame format
- MAC Addressing

- Flow frames are transmitted over Ethernet links
 - Error detection
-

MAC Address

⇒ Media Access Control address is an identifier used at the link layer.

⇒ Written as six hexadecimal octets

3c : 52 : 82 : A1 : 4f : 09

12 hex digit $\times$ 4 bits

= 48 bits

Unicast

A — switch — B

Dest M_{ac} = MAC-B

Source MAC = MAC-A.

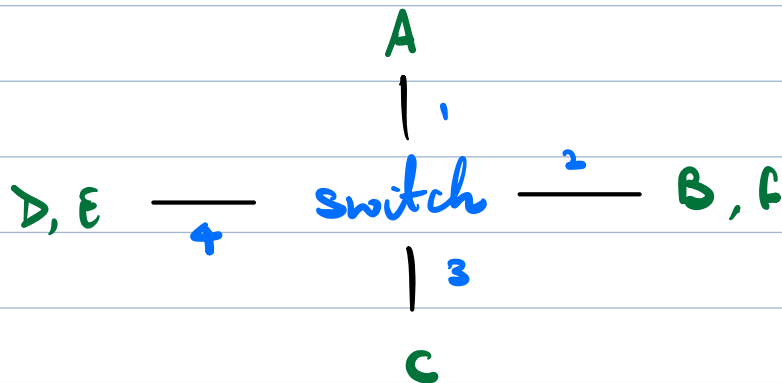
Broadcast

⇒ Every device on local layer-2 broadcast domain

⇒ Special broadcast MAC Address
FF: FF: FF: FF: FF: FF

MultiCast

A — selected group.



MAC Address Table / forwarding Table.